

Results - Lab13

```
[root@iZwz989thjg5p4g6ibyphsZ Lab13]# psql "postgresql://admin:000000@127.0.0.1:5432/lab13" -f './test1.sql'
```

departure	arrival	day_op	dep_time	carrier	airline	flightnum	duration	aircraft
AMS	DTW	1234567	07:55	DL	DL	251	505	777
AMS	DTW	1234567	07:55	DL	KL	6051	505	777
ATL	MAN	3457	18:55	DL	AF	3677	505	76W
ATL	MAN	3457	18:55	DL	DL	64	505	76W
ATL	MAN	3457	18:55	DL	KL	6164	505	76W
ATL	MAN	3457	18:55	DL	VS	4024	505	76W
BOM	ZRH	1234567	01:20	LX	LX	155	505	333
CAN	DOH	7	00:55	QR	QR	1875	505	77W
CDG	ORD	1234567	11:40	UA	LH	9334	505	763
CDG	ORD	1234567	11:40	UA	UA	986	505	763
CDG	ORD	12567	12:40	US	US	6679	505	763
CLT	MAD	12357	16:45	US	US	748	505	767
CPH	EWR	346	18:35	SK	SK	901	505	343
CPH	YYZ	24567	12:25	LH	LH	5508	505	763
DEL	FRA	1234567	03:05	LH	LH	761	505	74H
DUS	ORD	1	12:45	LH	LH	436	505	343
DUS	ORD	1	12:45	LH	UA	9259	505	343
DUS	ORD	12367	12:45	LH	LH	436	505	343
DUS	ORD	12367	12:45	LH	UA	9259	505	343
DUS	YYZ	12367	12:25	LH	AC	9174	505	343
DUS	YYZ	12367	12:25	LH	LH	472	505	343
FCO	BOS	13457	15:20	AZ	AZ	614	505	330
FCO	BOS	13457	15:20	AZ	DL	6587	505	330
FRA	YYZ	1234567	10:00	AC	AC	873	505	77W
FRA	YYZ	1234567	10:00	AC	LH	6676	505	77W
FRA	YYZ	12	17:00	AC	AC	877	505	77W
FRA	YYZ	1234567	17:00	AC	LH	5466	505	333
FRA	YYZ	12	17:00	AC	LH	6780	505	77W
FRA	YYZ	1234567	17:00	AC	AC	877	505	77W
FRA	YYZ	1234567	17:00	AC	LH	6780	505	77W
GRU	MTA	1234567	00:10	AA	AA	930	505	777

```
[root@iZwz989thjg5p4g6ibyphsZ Lab13]# psql "postgresql://admin:000000@127.0.0.1:5432/lab13" -f './test2.sql'
```

QUERY PLAN

Seq Scan on movies (cost=0.00..164.04 rows=9204 width=31)
(1 row)

QUERY PLAN

Seq Scan on movies (cost=0.00..187.05 rows=6000 width=31)
Filter: (movieid < 6000)
(2 rows)

QUERY PLAN

Index Scan using movies_pkey on movies (cost=0.29..12.75 rows=198 width=31)
Index Cond: (movieid < 200)
(2 rows)

```
[root@iZwz989thjg5p4g6ibyphsZ Lab13]# psql "postgresql://admin:000000@127.0.0.1:5432/lab13" -f './test3.sql'
BEGIN

                                QUERY PLAN
-----
Index Scan using movies_pkey on movies  (cost=0.29..12.75 rows=198 width=31) (actual time=0.013..0.040 rows=199 loops=1)
  Index Cond: (movieid < 200)
Planning Time: 0.448 ms
Execution Time: 0.095 ms
(4 rows)

ROLLBACK
```

```
ROLLBACK
[root@iZwz989thjg5p4g6ibyphsZ Lab13]# psql "postgresql://admin:000000@127.0.0.1:5432/lab13" -f './test4.sql'
psql:test4.sql:2: NOTICE:  table "my_movies" does not exist, skipping
DROP TABLE
SELECT 9204
psql:test4.sql:7: NOTICE:  index "idx_my_movies_year" does not exist, skipping
DROP INDEX
psql:test4.sql:8: NOTICE:  index "idx_my_movies_runtime" does not exist, skipping
DROP INDEX
CREATE INDEX
CREATE INDEX

                                QUERY PLAN
-----
Bitmap Heap Scan on my_movies  (cost=9.51..13.53 rows=1 width=242)
  Recheck Cond: ((runtime = 100) AND (year_released = 2000))
    -> BitmapAnd  (cost=9.51..9.51 rows=1 width=0)
      -> Bitmap Index Scan on idx_my_movies_runtime  (cost=0.00..4.63 rows=46 width=0)
            Index Cond: (runtime = 100)
      -> Bitmap Index Scan on idx_my_movies_year  (cost=0.00..4.63 rows=46 width=0)
            Index Cond: (year_released = 2000)
(7 rows)
```

```
[root@iZwz989thjg5p4g6ibyphsZ Lab13]# psql "postgresql://admin:000000@127.0.0.1:5432/lab13" -f './test4-2.sql'

                                QUERY PLAN
-----
Seq Scan on flights  (cost=0.00..1549.36 rows=1616 width=36)
  Filter: (duration = 105)
(2 rows)
```

```
[root@iZwz989thjg5p4g6ibyphsZ Lab13]# psql "postgresql://admin:000000@127.0.0.1:5432/lab13" -f './test5.sql'

                                QUERY PLAN
-----
Nested Loop  (cost=0.00..193.36 rows=200 width=49)
  -> Seq Scan on countries c2  (cost=0.00..4.31 rows=1 width=18)
      Filter: (country_code = 'cn'::bpchar)
  -> Seq Scan on movies m  (cost=0.00..187.05 rows=200 width=31)
      Filter: (country = 'cn'::bpchar)
(5 rows)
```

```
[root@iZwz989thjg5p4g6ibyphsZ Lab13]#
```

```
[root@iZwz989thjg5p4g6ibyphsZ Lab13]# psql "postgresql://admin:000000@127.0.0.1:5432/lab13" -f './test6.sql'

                                QUERY PLAN
-----
Nested Loop  (cost=0.57..664.03 rows=187 width=41)
  -> Index Scan using movies_pkey on movies m  (cost=0.29..12.75 rows=198 width=31)
      Index Cond: (movieid < 200)
  -> Index Only Scan using credits_pkey on credits c  (cost=0.29..3.28 rows=1 width=10)
      Index Cond: ((movieid = m.movieid) AND (credited_as = 'D'::bpchar))
(5 rows)
```

```
[root@iZwz989thjg5p4g6ibyphsZ Lab13]# psql "postgresql://admin:000000@127.0.0.1:5432/lab13" -f './test7.sql'
DROP INDEX
CREATE INDEX
```

QUERY PLAN

```
-----
Index Scan using movies_multi_index on my_movies  (cost=0.29..8.30 rows=1 width=31)
  Index Cond: (movieid = 20)
(2 rows)
```

QUERY PLAN

```
-----
Index Scan using movies_multi_index on my_movies  (cost=0.29..9.28 rows=1 width=31)
  Index Cond: ((movieid < 100) AND (year_released = 2000))
(2 rows)
```

QUERY PLAN

```
-----
Bitmap Heap Scan on my_movies  (cost=5.20..81.40 rows=118 width=31)
  Recheck Cond: (year_released = 2000)
  -> Bitmap Index Scan on idx_my_movies_year  (cost=0.00..5.17 rows=118 width=0)
       Index Cond: (year_released = 2000)
(4 rows)
```

QUERY PLAN

```
-----
Bitmap Heap Scan on my_movies  (cost=4.32..17.54 rows=4 width=31)
  Recheck Cond: (runtime = 200)
  -> Bitmap Index Scan on idx_my_movies_runtime  (cost=0.00..4.32 rows=4 width=0)
       Index Cond: (runtime = 200)
(4 rows)
```

```
[root@iZwz989thjg5p4g6ibyphsZ Lab13]# psql "postgresql://admin:000000@127.0.0.1:5432/lab13" -f './test8.sql'
psql:test8.sql:1: NOTICE:  index "movies_title_index" does not exist, skipping
DROP INDEX
CREATE INDEX
```

QUERY PLAN

```
-----
Bitmap Heap Scan on my_movies  (cost=4.30..11.33 rows=2 width=31)
  Recheck Cond: ((title)::text = 'Armaan'::text)
  -> Bitmap Index Scan on movies_title_index  (cost=0.00..4.30 rows=2 width=0)
       Index Cond: ((title)::text = 'Armaan'::text)
(4 rows)
```

QUERY PLAN

```
-----
Seq Scan on my_movies  (cost=0.00..210.06 rows=46 width=31)
  Filter: (upper((title)::text) = 'ARMAAN'::text)
(2 rows)
```

```
psql:test8.sql:7: NOTICE:  index "movies_upper_title_index" does not exist, skipping
DROP INDEX
CREATE INDEX
```

QUERY PLAN

```
-----
Bitmap Heap Scan on my_movies  (cost=4.64..72.12 rows=46 width=31)
  Recheck Cond: (upper((title)::text) = 'ARMAAN'::text)
  -> Bitmap Index Scan on movies_upper_title_index  (cost=0.00..4.63 rows=46 width=0)
       Index Cond: (upper((title)::text) = 'ARMAAN'::text)
(4 rows)
```